

SAFETY INSPECTION REPORT

Report No. : SIR-2021-034
Inspection Type : Periodic Safety Inspection ? Bay 4, Zone C
Date : 16 July 2021
Inspector : Priya Nair, HSE Manager
Accompanied By : Anil Mehta (I&E Technician), Rajiv Sharma (Maintenance Supervisor)
Next Inspection : January 2022

1. SCOPE

Periodic inspection of Bay 4, Zone C covering fixed gas detection system, hot work permit procedures, and general housekeeping. Inspection triggered by 18-month post-incident review schedule following NMR-2019-047 (March 2019 near-miss involving GB-14).

2. FIXED GAS DETECTION ? GB-14

Equipment Tag : GB-14 (Draeger Polytron 2 XP, S/N: 7729-4412)
Location : Bay 4, Zone C, Elevation 3.2m
Installation : March 2017

2.1 Calibration Records Review

Last calibration date: 17 March 2021 (WO-4198, Anil Mehta)
Due date: September 2021 ? (within schedule)
Pre-calibration reading: +0.8% LEL drift (ACCEPTABLE ? within ±1% tolerance)
Span calibration result: 50.2% LEL at 50% reference. PASS.

2.2 Physical Inspection

Housing: No damage, IP65 integrity intact.
Cable gland: Secure, no corrosion.
Display: Functioning. Current reading: 0.0% LEL.
Alarm horn: Tested ? audible at 25 metres. PASS.
Sensor age: 4 years 4 months (installation March 2017). Sensor element last replaced: 14 February 2019 (WO-3847). Age: 29 months.

? OBSERVATION: Sensor element is 5 months past Draeger's 24-month replacement recommendation. Work order recommended (see Action Items).

2.3 Alarm Escalation Procedure

Procedure SHP-04 (Shift Handover and Alarm Escalation) reviewed.
Shift operators interviewed: Suresh Kumar, Vijay Patil.
Both demonstrated correct knowledge of 15% LEL notification threshold and 25% LEL immediate escalation + hot work suspension requirement.
[Note: Following NMR-2019-047, CA-03 was implemented ? refresher training for Suresh Kumar completed March 2019. Records sighted and satisfactory.]

3. HOT WORK PERMIT SYSTEM ? BAY 4

3.1 Procedure Review

Current procedure: HWP-PROC-2019-Rev2 (updated post NMR-2019-047).
Key update from 2019 revision: Requires mandatory gas reading from nearest fixed detector (GB-14 for Bay 4 work) within 30 minutes of permit issuance.
This requirement (CA-02 from NMR-2019-047) is correctly incorporated. ?

3.2 Permit Records Audit (last 6 months: Jan-Jul 2021)

12 hot work permits reviewed for Bay 4.
Gas reading field compliance: 10/12 permits had GB-14 reading recorded.
2 permits (HWP-2021-044, HWP-2021-067): Gas reading field marked "0% ? area clear" without time recorded. Non-compliant with procedure (time of reading must be recorded).

? FINDING: Partial non-compliance ? 2/12 permits (17%) missing reading timestamp.

4. CORRECTIVE ACTIONS